

NORTH SYDNEY GIRLS' HIGH SCHOOL

TRIAL HIGHER SCHOOL CERTIFICATE, 1990

MATHEMATICS 3U/4U COMMON PAPER

QUESTION 1

(a) Simplify $\frac{\tan 5x - \tan x}{1 + \tan 5x \cdot \tan x}$

- (b) Find the acute angle between the lines $2y - x + 1 = 0$ and $y = 5x + 2$ (give answer correct to the nearest minute).

- (c) If α, β, γ are the roots of the equation $2x^3 + 5x - 3 = 0$, find the value of $\alpha^{-1} + \beta^{-1} + \gamma^{-1}$.

- (d) Find the co-ordinates of the point that divides the interval PQ externally in the ratio 3 : 4 if P is the point (2, 5) and Q is the point (-1, 0).

- (e) Find the limiting sum of $1 + \sin^2 x + \sin^4 x + \dots$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$

QUESTION 2

- (a) A circular oil slick lies on the surface of calm water. Its area is increasing at the rate of $12 \text{ m}^2/\text{min}$. At what rate is the radius increasing at the time at which the radius is 3 metres?

- (b) Use the substitution $u = \sin x$ to show that

$$\int_0^{\frac{\pi}{6}} \frac{\cos x \cdot dx}{4 \sin^2 x + 1} = \frac{\pi}{8}$$

- (c) PQR is an equilateral triangle. QR is produced to T so that $RT = \frac{1}{3} QR$.

If $RT = x$ units, prove that $PT = x\sqrt{13}$ units.

- (d) Find the derivative, with respect to x , of:

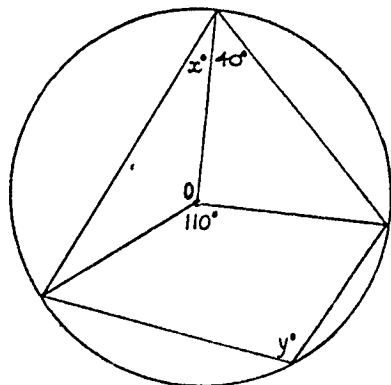
(i) $\log(x^2 + 1)$

(ii) $e^{x^2} \cos 4x$

QUESTION 3

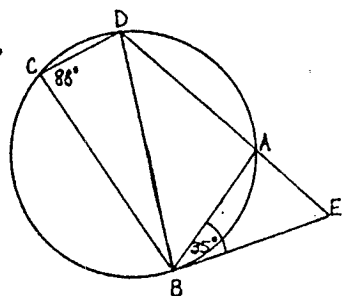
- (a) (i) Find the value of the pronumerals x and y , giving reasons for your answers.

(O is the centre of the circle)



- (ii) If $\angle BCD = 88^\circ$ and $\angle EBA = 35^\circ$, find $\angle BAE$ and $\angle BDE$, giving reasons for your answers.

(BE is a tangent)



- (b) Find all values of θ in the range $0 \leq \theta \leq 360^\circ$ for which

$$3 \cos \theta + \sqrt{3} \sin \theta = \sqrt{3}$$

- (c) Solve the equation $x^{\frac{4}{3}} - 2x^{\frac{1}{3}} - 4 = 0$

Expand and simplify your values of x , leave as surds.

QUESTION 4

- (a) The equation $x^2 = 1 - x$ has approximately the solution $x = 0.5$. Use one application of Newton's Method to obtain a better approximation.

- (b) Evaluate exactly $\int_0^1 \left(e^{-x} + \frac{1}{1+x} + \frac{1}{\sqrt{1-x^2}} \right) dx$

- (c) In how many ways can a train of ten carriages be arranged if four of the carriages

- (i) are to be kept in a given order? |||| |||| ||||

(ii) must be kept together but in any order? (111) |||| ||||

- (d) If $y = \left(\frac{e}{2}\right)^x$ show that $\frac{1}{y} \cdot \frac{dy}{dx} = 1 - \log_e 2$

- (e) Find the limit of $\frac{\sin 4h}{\tan 5h}$ as h approaches 0.

QUESTION 4

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- (e) Find the limit of $\frac{\sin 4h}{\tan 5h}$ as h approaches 0.

QUESTION 5

(a) Solve $\frac{x^2 + 6}{x} < 5$

$0.6x^2$
 $\frac{1}{2}$

$$\frac{1 \pm \sqrt{1 - 4(0.8)(0.6)}}{2(0.5)}$$

$0.5x^2 + 0.5 - x = 0$

(b) Consider $f(x) = \frac{x}{x^2 + 1}$

$0.9x^2 + 0.9 - x = 0$

$$\frac{1 \pm \sqrt{1 - 4(0.9)(0.9)}}{2(0.9)}$$

- (i) Prove that $f(x)$ is an odd function.
- (ii) Find the co-ordinates and nature of its turning points.
- (iii) Find the range of the function.
- (iv) Hence or otherwise, sketch $y = f(x)$.
- (v) Find the area enclosed by the curve, the x -axis and the lines $x = -1$ and $x = 1$.

QUESTION 6

- (a) The rate at which an object changes temperature is proportional to the difference between its temperature and that of the surrounding medium, that is:

$$\frac{dT}{dt} = -k(T - M)$$

where T is the temperature at any time t and M is the temperature of the surrounding medium (a constant).

- (i) Show that the temperature, T , of the body at any time t is given by the formula

$$T = M + Ae^{-kt}$$

- (ii) A metal bar has a temperature of 1230°C and cools to 1030°C in 10 minutes, when the surrounding temperature is 30°C . How long will it take to cool to 80°C ?

- (b) The tangent at $P(2ap, ap^2)$ on the parabola $x^2 = 4ay$ meets the x -axis in T . The normal at P meets the y -axis in N .

- (i) Find the co-ordinates of M , the midpoint of TN .

- (ii) Show that the locus of M is the parabola $x^2 = \frac{a}{2}(y - a)$

7.

QUESTION 7

- (a) The coefficient of x in the expansion of $\left(x + \frac{1}{ax^2}\right)^7$ is $\frac{7}{3}$.

Find all the possible values of 'a'.

- (b) Sketch the graph of $y = 4 \sin^{-1}(2x + 1)$, stating its largest possible domain and range.

- (c) Prove by Mathematic Induction that

$$3^n > 1 + 2n \quad \text{for } n \geq 0$$

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Question 1

(a) $\frac{\tan 5x - \tan x}{1 + \tan 5x \tan x}$ let $A = 5x$
 $B = x$
 Now $\tan(A-B) = \frac{\tan A - \tan B}{1 + \tan A \tan B}$
 $\therefore \frac{\tan 5x - \tan x}{1 + \tan 5x \tan x} = \tan(5x - x)$
 $= \tan 4x$ ✓

(b) $2y - x + 1 = 0$ $y = 5x + 2$
 $y = \frac{x+1}{2}$ $\frac{dy}{dx} = 5$ ✓
 $\therefore \frac{dy}{dx} = \frac{1}{2}$ ✓
 $\tan \theta = \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right|$
 $= \left| \frac{\frac{1}{2} - 5}{1 + \frac{1}{2} \times 5} \right|$
 $\theta = 52^\circ 8'$ ✓

(c) $2x^3 + 5x - 3 = 0$ α, β, γ are roots
 let $y = \frac{1}{x}$ $x = \frac{1}{y}$ sub in eqⁿ
 $\frac{2}{y^3} + \frac{5}{y} - 3 = 0$ multiply by y^3
 $2 + 5y^2 - 3y^3 = 0$ ✓
 $\therefore$ req'd eqⁿ is $3x^3 - 5x^2 - 2 = 0$
 (when reverted back to x)

(d) $x = \left(\frac{mx_2 + nx_1}{m+n} \right) = \frac{3(0) + 4(2)}{3+4}$ Question says
 $= \frac{8}{7}$ External division
 $y = \frac{my_2 + ny_1}{m+n} = \frac{3(0) + 4(5)}{3+4}$
 $= \frac{20}{7}$ $\therefore P\left(\frac{8}{7}, \frac{20}{7}\right)$

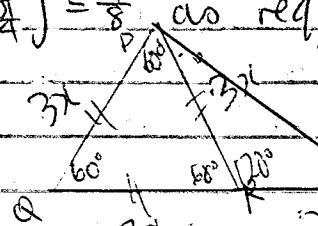
(e) $1 + \sin^2 x + \sin^4 x + \dots$ for $-\frac{\pi}{2} < x < \frac{\pi}{2}$
 $a = 1$ $r = \sin^2 x$
 $S_{\infty} = \frac{a}{1-r} = \frac{1}{1-\sin^2 x} = \frac{1}{\cos^2 x} = \sec^2 x$

Question 2

(a) $\frac{dA}{dt} = 12 \text{ m}^2/\text{min}$ ✓
 Find $\frac{dr}{dt}$ When $r = 3 \text{ m}$
 circular oil slick = sphere? If so,
 $A = \pi r^2$ $\frac{dA}{dr} = 2\pi r$
 $\frac{dr}{dt} = \frac{dA}{dt} \cdot \frac{dr}{dA}$ $\frac{dr}{dt} = \frac{1}{2\pi r}$
 $= 12 \times \frac{1}{2\pi} = \frac{6}{\pi} \text{ m/min}$

(b) $\int_0^{\frac{\pi}{2}} \frac{\cos x dx}{4\sin^2 x + 1}$ $= \frac{\pi}{8}$ $u = \sin x$
 $du = \cos x dx$
 $\int_0^{\frac{\pi}{2}} \frac{du}{4u^2 + 1}$ ✓

$\frac{1}{4} \int_0^{0.5} \frac{du}{u^2 + \frac{1}{4}}$
 $= \frac{1}{4} \times 2 \left[\tan^{-1} 2u \right]_0^{0.5}$ ✓
 $= \frac{1}{2} \left[\frac{\pi}{4} \right] = \frac{\pi}{8}$ as req'd

(c)  Prove $PT = \sqrt{3}x$

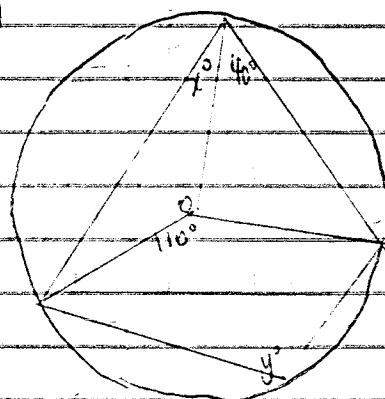
$(PT)^2 = (3x)^2 + x^2 - 2(3x)(x)\cos 120^\circ$
 $(PT)^2 = 10x^2 - 6x^2 \cos 120^\circ$
 $= 10x^2 - 6x^2 \left(-\frac{1}{2}\right)$ ✓
 $= 13x^2$
 $\therefore PT = \sqrt{13}x$ units ✓ as req'd

(d) i) $\log(x^3 + 1)$ let $u = x^3 + 1$
 $\frac{dy}{dx} = \frac{dy}{du} \cdot \frac{du}{dx}$ $\frac{du}{dx} = 3x^2$
 $= \frac{1}{u} \times 3x^2$
 $= \frac{3x^2}{x^3 + 1}$ ✓

ii) $e^{x^2} \cos 4x$
 $= e^{x^2} \frac{\sin 4x}{4} + \cos 4x (2xe^{x^2})$ ✓
 $= e^{x^2} \left[\frac{\sin 4x}{4} + 2x \cos 4x \right]$ ✓

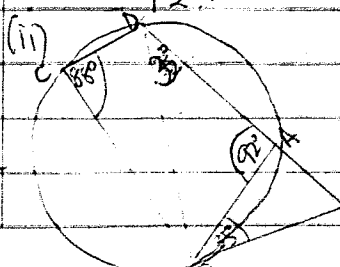
Question 3

(a) i)



$x + 40 = \frac{110}{2}$ ($\angle$ at centre ✓
 twice angle at circumference)
 $\therefore x = 15^\circ$ ✓

$y = 180 - 40 - 15$ opp $\angle$ s of cyclic quad are supplementary
 $= 125^\circ$



$\angle BDE = 35^\circ$ ($\angle$ in alt. segment
 $\angle DAB = 180 - 88 = 92^\circ$
 opp $\angle$ of cyclic quad = 180
 $\angle BAE = 88^\circ$ ✓

for same coeff. choose

$R \cos(\theta - \alpha)$ method is easier $\Rightarrow 2\sqrt{3} \cos(\theta - \frac{\pi}{6}) = \sqrt{3}$
 $\cos(\theta - \frac{\pi}{6}) = \frac{1}{2}$ for $-\pi \leq \theta - \frac{\pi}{6} \leq \pi$

(b) $0 \leq x \leq 360^\circ$, $3\cos\theta + \sqrt{3}\sin\theta = \sqrt{3}$

let $\tan\frac{\theta}{2} = t$ $\sin\theta = \frac{2t}{1+t^2}$ $\cos\theta = \frac{1-t^2}{1+t^2}$

$3(\frac{1-t^2}{1+t^2}) + \sqrt{3}(\frac{2t}{1+t^2}) = \sqrt{3}$ ✓

$3 - 3t^2 + 2\sqrt{3}t = \sqrt{3} + \sqrt{3}t^2$

$(\sqrt{3}+3)t^2 - 2\sqrt{3}t + \sqrt{3}-3 = 0$

$t = \frac{2\sqrt{3} \pm \sqrt{12 - 4(\sqrt{3}+3)(\sqrt{3}-3)}}{2(\sqrt{3}+3)}$

$= \frac{2\sqrt{3} \pm \sqrt{12 - 4(3-9)}}{2\sqrt{3}+6}$

$= \frac{2\sqrt{3} \pm 6}{2(\sqrt{3}+3)} = \frac{\sqrt{3} \pm 3}{\sqrt{3}+3} = 1 \text{ or } \frac{\sqrt{3}-3}{\sqrt{3}+3}$

Now $\frac{\sqrt{3}-3}{\sqrt{3}+3} \times \frac{\sqrt{3}-3}{\sqrt{3}-3} = \frac{3-6\sqrt{3}+9}{3-9}$

$= \frac{12-6\sqrt{3}}{-6} = -2 + \sqrt{3}$

$\therefore \tan\frac{\theta}{2} = \sqrt{3}-2 \text{ or } 1$

$\frac{\theta}{2} = n\pi + \tan^{-1}(1) \text{ or } \frac{\theta}{2} = n\pi + \tan^{-1}(\sqrt{3}-2)$

$\theta = \frac{\pi}{2}, \frac{11\pi}{6}, 90^\circ, 330^\circ$

(c) $x^{\frac{2}{3}} - 2x^{\frac{1}{3}} - 4 = 0$

let $u = x^{\frac{1}{3}}$

$u^2 - 2u - 4 = 0$

$u = \frac{2 \pm \sqrt{4 - 4(-4)}}{2} = \frac{2 \pm 2\sqrt{5}}{2} = 1 \pm \sqrt{5}$ ✓

$\therefore x^{\frac{1}{3}} = 1 \pm \sqrt{5}$

$x = 16 + 8\sqrt{5} \text{ or } 16 - 8\sqrt{5}$

Question 4.

(a) $x^2 = 1 - x$ root near $x_1 = 0.5$

$x^2 + x - 1 = 0$

$\therefore x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$ ✓

$f(0.5) = -\frac{1}{4}$ ✓

$f'(x) = 2x + 1$

$f'(0.5) = 2$

$x_2 = 0.5 - \frac{-0.25}{2}$ ✓

$\therefore x_2 = 0.625$ ✓

(b) $\int_0^1 (e^{-x} + \frac{1}{1+x} + \frac{1}{\sqrt{1-x^2}}) dx$

$= [-e^{-x} + \ln(1+x) + \sin^{-1}x]_0^1$

$= [-\frac{1}{e} + \ln 2 + \frac{\pi}{2}] - [-1 + 0 + 0]$

$= 1 - \frac{1}{e} + \ln 2 + \frac{\pi}{2}$

(c) (i) $6!$

(d) $y = (\frac{e}{2})^x$ show $\frac{1}{y} \frac{dy}{dx} = 1 - \ln 2$

$y = \frac{e^x}{2^x}$
 $\frac{dy}{dx} = \frac{e^x \cdot 2^x - e^x \cdot 2^x \ln 2}{2^{2x}}$
 $= \frac{e^x (1 - \ln 2)}{2^{2x}} = \frac{e^x (1 - \ln 2)}{2^x}$ ✓

Now LHS = $\frac{1}{y} \frac{dy}{dx}$

$= \frac{2^x}{e^x} \times \frac{e^x}{2^x} (1 - \ln 2) = 1 - \ln 2 = \text{RHS}$ ✓

(e) Find the limit of $\frac{\sinh x}{\tanh x}$ as $x \rightarrow \infty$

$\lim_{x \rightarrow \infty} \frac{f(x)}{g(x)} = \frac{L}{m}$ where L is limit of $f(x)$ & m is limit of $g(x)$

$\therefore \lim_{h \rightarrow \infty} \sinh h = \frac{e^h - e^{-h}}{2} \approx \frac{e^h}{2}$

$\lim_{h \rightarrow \infty} \tanh h = \frac{e^h - e^{-h}}{e^h + e^{-h}} \approx \frac{e^h}{e^h} = 1$

$\therefore \lim_{h \rightarrow \infty} \frac{\sinh h}{\tanh h} = \frac{e^h/2}{1} = \frac{e^h}{2}$ ✓

Question 5:

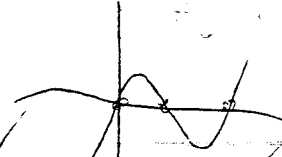
(a) $\frac{x^2+6}{x} < 5$ multiply by x^2

$x(x^2+6) < 5x^2$

$x^3 + 6x - 5x^2 < 0$

$x(x^2 - 5x + 6) < 0$ ✓

$x(x-2)(x-3) < 0$ ✓



$\therefore x < 0$ ✓, $2 < x < 3$ ✓

(b) (i) $f(x) = \frac{x}{x^2+1}$

$f(-x) = \frac{-x}{x^2+1} = -f(x)$ ✓

$\therefore$ it's an odd fn ✓

(ii) $f'(x) = \frac{x^2+1-2x^2}{(x^2+1)^2} = \frac{-x^2+1}{(x^2+1)^2}$ ✓

$= \frac{1}{x^2+1} - \frac{2x^2}{(x^2+1)^2}$

(i) $f'(x) = 0$ at $x^2 = 1 \Rightarrow x = \pm 1$

$-x^2 + 1 = 0 \therefore$ turning point at $(1, \frac{1}{2})$

$\frac{d^2y}{dx^2} = (x^2+1)^{-2}(-2x) - (-x^2+1)(-2x)(x^2+1)^{-3}$

IF $\frac{d^2y}{dx^2}$ is difficult find $(x^2+1)^4$

use

$1 - x = 1^+ \quad 1^+$

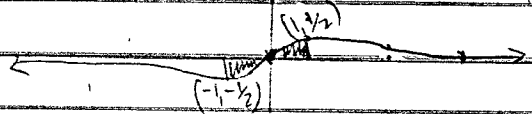
$\frac{-2x(x^2+1)^3 - 4x(x^2+1)(-x^2+1)}{(x^2+1)^4} = \frac{-2x(x^2+1)^2 - 4x(-x^2+1)}{(x^2+1)^3}$

concave down.

(ii) $x \neq 1 \leq y \leq \frac{1}{2}$ ✓

(iv)

$f(x) = \frac{x}{x^2+1}$ ✓



(v) $A = \int_{-1}^1 \frac{x}{x^2+1} dx$ let $u = x^2+1$
 $\frac{du}{dx} = 2x \Rightarrow dx = \frac{du}{2x}$
 $= \frac{1}{2} \int \frac{du}{u} = \frac{1}{2} \ln(x^2+1) \Big|_{-1}^1$
 $= \frac{1}{2} \ln 2 + \frac{1}{2} \ln 2 = \ln 2$

Question 6.

(a) i) $T = M + Ae^{-kt} \therefore Ae^{-kt} = T - M$ ✓
 $\frac{dT}{dt} = -kAe^{-kt} = -k(T - M)$ as req'd

(ii) $M = 30$ when $t = 0$ $T = 1230$

$1230 = 30 + A \therefore A = 1200$ ✓

when $t = 10$ $T = 1030$

$1030 = 30 + 1200e^{-k \cdot 10}$

$\therefore k = \frac{\ln \frac{1000}{1200}}{-10} = 0.018232155$

when $T = 80$ $t = ?$

$80 = 30 + 1200e^{-kt}$ ✓

$t = \frac{\ln \frac{1}{24}}{-0.018232155} = 174 \text{ min } 19 \text{ seconds}$

(b) i) $x^2 = 4ay$ $P(2ap, ap^2)$ should draw a diagram.

$\frac{dy}{dx} = \frac{x}{2a} = p$

$\therefore$ eqⁿ of tangent at P

$(y - ap^2) = p(x - 2ap)$

$u = px - ap^2 \quad T(x, 0)$

$\therefore x = ap \quad \therefore T(ap, 0)$ ✓

$m_{\text{normal}} = -\frac{1}{p}$

eqⁿ of normal $(y - ap^2) = -\frac{1}{p}(x - 2ap)$

$x + py = 2ap + ap^3$

$N(0, y) \therefore y = 2a + ap^2$ ✓

$\therefore$ midpoint TN $x = \frac{ap+0}{2} = \frac{ap}{2}$

$y = \frac{a}{2}(2 + p^2)$ ✓

$M(\frac{ap}{2}, \frac{a}{2}(p^2+2))$

(ii) $x = \frac{ap}{2} \quad p = \frac{2x}{a}$ ①

$y = \frac{a}{2}(p^2+2)$ ② sub ① in ②

$y = \frac{a}{2}(\frac{4x^2}{a^2} + 2)$
 $= \frac{2x^2}{a} + a$ ✓

$(y-a)\frac{a}{2} = x^2$

$\therefore$ locus of M is parabola

$x^2 = \frac{a}{2}(y-a)$ ✓

Question 7.

(a) -

(b) $y = 4 \sin^{-1}(2x+1)$

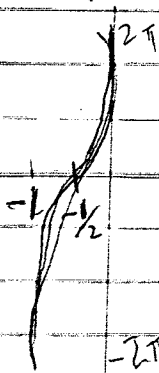
$0 \leq 2x+1 \leq 1$

$= \{-2 \leq 2x \leq 0\}$

$= \{-1 \leq x \leq 0\}$ ✓

Ry : $\{-\frac{\pi}{2} \leq \frac{y}{4} \leq \frac{\pi}{2}\}$

$\{-2\pi \leq y \leq 2\pi\}$ ✓



(c) Step 1: Prove true for $n=1$

$$3 \geq 1+2 \quad \text{true for } n=1$$

Step 2: Assume true for $n=k$

$$\text{i.e. } 3^k \geq 1+2k$$

Step 3: Prove true for $n=k+1$

$$\text{u. RTP } 3^{k+1} \geq 1+2(k+1) = 3+2k$$

$$3 \cdot 3^k \geq 3(1+2k) \quad (\text{from assumption})$$

$$\text{Now since } 3+6k \geq 3+2k$$

$$\therefore 3^{k+1} \geq 1+2(k+1) \quad \checkmark$$

Step 4: since true for $n=1$

& assumed true for $n=k$

& proven true for $n=k+1$

$\therefore$ true for $n=1+1=2$

so on for all positive integers, n

Test true for inequality $n=2$

$$\therefore \text{LHS} = 3^2 = 9$$

$$\text{RHS} = 5$$

$$\therefore 3^2 > 1+2 \cdot 2$$